

Exercise 7

```
[ec2-user@ip-172-31-20-184 lab8]$ mpicc -o pi_bsend pi_bsend.c -lm
mpirun --oversubscribe -np 4 ./pi_bsend
Rank 0: trials = 2500000, count = 1962662
Root received count = 1962547 from rank 1
Root received count = 1963912 from rank 2
Root received count = 1964200 from rank 3

[Pi Bsend] Processes = 4
[Pi Bsend] Trials = 10000000
[Pi Bsend] Points inside = 7853321
[Pi Bsend] Estimate of pi = 3.1413284000
[Pi Bsend] Error = 0.0002642536
[Pi Bsend] Time = 0.272962 sec
[ec2-user@ip-172-31-20-184 lab8]$ |
```

In `pi_bsend.c` the workers send their count with `MPI_Bsend` instead of `MPI_Send`. Each worker attaches a buffer of `MPI_Pack_size(1, MPI_LONG_LONG) + MPI_BSEND_OVERHEAD` bytes with `MPI_Buffer_attach`, calls `MPI_Bsend` and then `MPI_Buffer_detach`, which waits until the message has left the buffer. Root is unchanged and still receives with `MPI_ANY_SOURCE` (in this run the counts happened to arrive in rank order). The result is the same again ($\pi = 3.1413284$).

Processes	Exercise 3 (MPI_Send)	Exercise 6 (ANY_SOURCE)	Exercise 7 (MPI_Bsend)
1	351.41 ms	352.58 ms	352.00 ms
2	263.77 ms	264.41 ms	264.34 ms
3	270.82 ms	269.19 ms	276.64 ms
4	273.35 ms	271.87 ms	300.79 ms
5	269.84 ms	276.27 ms	276.18 ms
6	273.56 ms	274.13 ms	278.07 ms
7	270.86 ms	270.49 ms	272.08 ms
8	252.59 ms	261.72 ms	253.72 ms

Buffered send made no real difference to the time either. The only larger difference from Exercise 3 is at 4 processes (300.79 ms against 273.35 ms, 10 %), and it does not appear at the other process counts.

MPI_Bsend returns as soon as the message is copied into the buffer, so a worker never waits for root, but the message is only one long long and MPI_Send already sends such a small message without waiting.

MPI_Bsend is useful when the message is large, or when the sender must continue before the receiver is ready.